



PCA Calculation

Q] Given the following data, use PCA to reduce the dimension from 2 to 1.

x	y
4	11
8	4
13	5
7	14

Step 1: Dataset Information

No. of features = 2
No. of observations = 4

Step 2: Computation of Mean of variables

$$\bar{x} = \frac{4 + 8 + 13 + 7}{4} = 8$$

$$\bar{y} = \frac{11 + 4 + 5 + 14}{4} = 8.5$$

Step 3: Computation of Covariance Matrix.

① Ordered pairs of x and y are:
(x, x), (x, y), (y, x), (y, y)

$(\vec{x}, \vec{y})$ } So, if we have n = 30
n² ordered pairs.
i.e. 900

x	y
4	11
8	4
13	5
7	14

② Covariance of all ordered pairs:

wkt,
$$Cov(x, x) = \frac{1}{N} \sum (x_{ik} - \bar{x}_i)(x_{jk} - \bar{x}_j) \longrightarrow \textcircled{1}$$





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$(x, x), (x, y), (y, x), (y, y)$

(x, y)

So, if we have $n=30$
 n^2 ordered pairs.
i.e. 900

x	y
4	11
8	4
13	5
7	14

$$\bar{x} = 8 \quad \bar{y} = 8.5$$

$$N = 4$$

② Covariance of all ordered pairs:

$$\text{wkt, } \text{Cov}(x, x) = \frac{1}{N-1} \sum_{k=1}^N (x_{ik} - \bar{x}_i)(x_{jk} - \bar{x}_j) \longrightarrow \textcircled{1}$$

But if cov of x wrt x , the eqn is given by,

$$\text{Cov}(x, x) = \frac{1}{N-1} \sum_{k=1}^N (x_i - \bar{x})^2 \longrightarrow \textcircled{2}$$

So, the $\text{cov}(x, x)$ is given by,

$$\textcircled{2} \Rightarrow \text{Cov}(x, x) = \frac{1}{4-1} [(4-8)^2 + (8-8)^2 + (13-8)^2 + (7-8)^2]$$

$$\boxed{\text{Cov}(x, x) = 14}$$



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x	y
4	11
8	4
13	5
7	14

$$\bar{x} = 8 \quad \bar{y} = 8.5$$

Now cov(x, y):

$$\text{Eq (1)} \Rightarrow \text{cov}(x, y) = \frac{1}{4-1} \left[(4-8)(11-8.5) + (8-8)(4-8.5) + (13-8)(5-8.5) + (7-8)(14-8.5) \right]$$

$$\boxed{\text{cov}(x, y) = -11}$$

$$\therefore \text{cov}(y, x) = \text{cov}(x, y)$$

$$\therefore \boxed{\text{cov}(y, x) = -11}$$

Now, cov(y, y):

$$\text{Eq (2)} \Rightarrow \text{cov}(y, y) = \frac{1}{4-1} \left[(11-8.5)^2 + (4-8.5)^2 + (5-8.5)^2 + (14-8.5)^2 \right]$$

$$\boxed{\text{cov}(y, y) = 23}$$

x	y
4	11
8	4
13	5
7	14

$$\bar{x} = 8 \quad \bar{y} = 8.5$$

(3) Form Covariance Matrix of size $n \times n$:





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③ Form Covariance Matrix of size $n \times n$:

$$S = \begin{bmatrix} \text{Cov}(x, x) & \text{Cov}(x, y) \\ \text{Cov}(y, x) & \text{Cov}(y, y) \end{bmatrix}$$

$$\text{Cov. Matrix } S = \begin{bmatrix} 14 & -11 \\ -11 & 23 \end{bmatrix}$$

Step 4: Eigen value, Eigen vector, Normalized Eigen Vector.

① Eigen value:

$$\text{determinant}(S - \lambda I) = 0$$

$$\det \begin{bmatrix} 14 - \lambda & -11 \\ -11 & 23 - \lambda \end{bmatrix} = 0$$

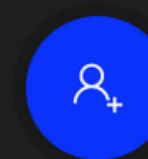
Wkt, S size is 2×2 .

So, I will be 2×2 .

$$I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\lambda I = \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix}$$





$$\det \begin{pmatrix} 14-\lambda & -11 \\ -11 & 23-\lambda \end{pmatrix} = 0$$

Note: To find determinant, cross multiplication.

$$(14-\lambda)(23-\lambda) - (-11 \times -11) = 0$$

$$\lambda^2 - 37\lambda + 201 = 0 \longrightarrow \textcircled{4}$$

wkt, Root of the quadratic eqn is given by,

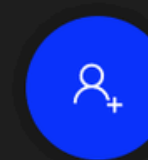
$$\frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

So, Here $a=1$, $b=-37$, $c=201$.

So, finally we get, $\lambda = 30.3849 \text{ (or) } 6.6151$

Now, arrange the λ value in descending order.





Now, arrange the λ' value in descending order.

$$\lambda_1 > \lambda_2 \dots$$

$$\text{So, here } \lambda_1 = 30.3849 \rightarrow \text{PC 1} \checkmark$$

$$\lambda_2 = 6.6151$$

So, we choose the highest value of λ and proceed further.

② Eigen vector of λ_1 :

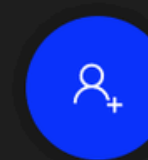
$$\text{Wkt, } (S - \lambda_1 I)U_1 = 0.$$

Here, we have to find out U_1 which is the Eigen vector of λ_1 .

$$\left[\begin{bmatrix} 14 & -11 \\ -11 & 23 \end{bmatrix} - \begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_1 \end{bmatrix} \right] \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} = 0$$

$$\begin{bmatrix} 14 - \lambda_1 & -11 \end{bmatrix} \begin{bmatrix} u_1 \end{bmatrix} = 0$$





$$\begin{bmatrix} 14 - \lambda_1 & -11 \\ -11 & 23 - \lambda_1 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} = 0$$

$$\begin{bmatrix} (14 - \lambda_1)u_1 - 11u_2 \\ -11u_1 + (23 - \lambda_1)u_2 \end{bmatrix} = 0$$

Now, from this we got 2 linear eqtns.

$$\text{i.e., } (14 - \lambda_1)u_1 - 11u_2 = 0 \rightarrow (5)$$

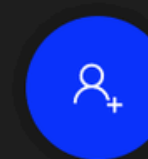
$$-11u_1 + (23 - \lambda_1)u_2 = 0 \rightarrow (6)$$

So, now we have to find out u_1 and u_2 from any of these eqtns.

$$\text{From (5), } (14 - \lambda_1)u_1 = 11u_2$$

$$\frac{u_1}{11} = \frac{u_2}{(14 - \lambda_1)} = 1$$





$$\frac{u_1}{11} = \frac{u_2}{(14-\lambda)} = 1$$

So, we get $u_1 = 11$, $u_2 = 14 - \lambda$

Eigen vector u_1 of $\lambda_1 = \begin{bmatrix} 11 \\ 14 - \lambda_1 \end{bmatrix}$

$$= \begin{bmatrix} 11 \\ 14 - 30.3849 \end{bmatrix} = \begin{bmatrix} 11 \\ -16.3849 \end{bmatrix}$$

③ Normalize the Eigen vector u_1 :

To normalize an eigen vector, divide the vector by its length.

$$e_1 = \begin{bmatrix} 11 / \sqrt{11^2 + (-16.3849)^2} \\ -16.3849 / \sqrt{11^2 + (-16.3849)^2} \end{bmatrix}$$





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$$e_1 = \begin{bmatrix} 11 / \sqrt{11^2 + (-16.3849)^2} \\ -16.3849 / \sqrt{11^2 + (-16.3849)^2} \end{bmatrix}$$

So, unit vector/Normalized eigen vector is given by,

For $\lambda_1 = 30.3849$ $\rightarrow e_1 = \begin{bmatrix} 0.5574 \\ -0.8303 \end{bmatrix}$ $\rightarrow$ This is nothing but for our 1st eigen value λ_1 's corresponding unit eigen vector.

Now, we have to find out for λ_2 's corresponding unit eigen vector.

For $\lambda_2 = 6.6151$ $\rightarrow e_2 = \begin{bmatrix} 0.8303 \\ 0.5574 \end{bmatrix}$

Step 5: Derive New dataset:

PC1
P_{11}
P_{12}

$\rightarrow$?? How to get these values?



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Step 5: Derive New dataset:

p_{c1}
p_{11}
p_{12}
p_{13}
p_{14}

?? How to get these values?

wkt, $p_{ij} = e_i^T \begin{bmatrix} x_{1j} - \bar{x}_1 \\ x_{2j} - \bar{x}_2 \\ x_{3j} - \bar{x}_3 \end{bmatrix}$

x	y
4	11
8	4
13	5
7	14

$\bar{x} = 8$ $\bar{y} = 8.5$

$p_{11} = e_i^T \begin{bmatrix} 4 - 8 \\ 11 - 8.5 \end{bmatrix}$
 $= \begin{bmatrix} 0.5574 & -0.8303 \end{bmatrix} \begin{bmatrix} -4 \\ 2.5 \end{bmatrix}$

$p_{11} = -4.3052$

x	y
4	11
8	4
13	5

$\begin{bmatrix} 8 - 8 \end{bmatrix}$



x	y
4	11
8 ✓	4 ✓
13	5
7	14

$\bar{x} = 8$ $\bar{y} = 8.5$

$$P_{12} = \begin{bmatrix} 0.5574 & -0.8303 \end{bmatrix} \begin{bmatrix} 8 - 8 \\ 4 - 8.5 \end{bmatrix}$$

$P_{12} = 3.7361$

In the same way, we get,

$P_{13} = 5.6928$

$P_{14} = -5.1238$

So now we got a 1D dataset from 2D.

P_{01}
-4.3052
3.7361
5.6928
-5.1238



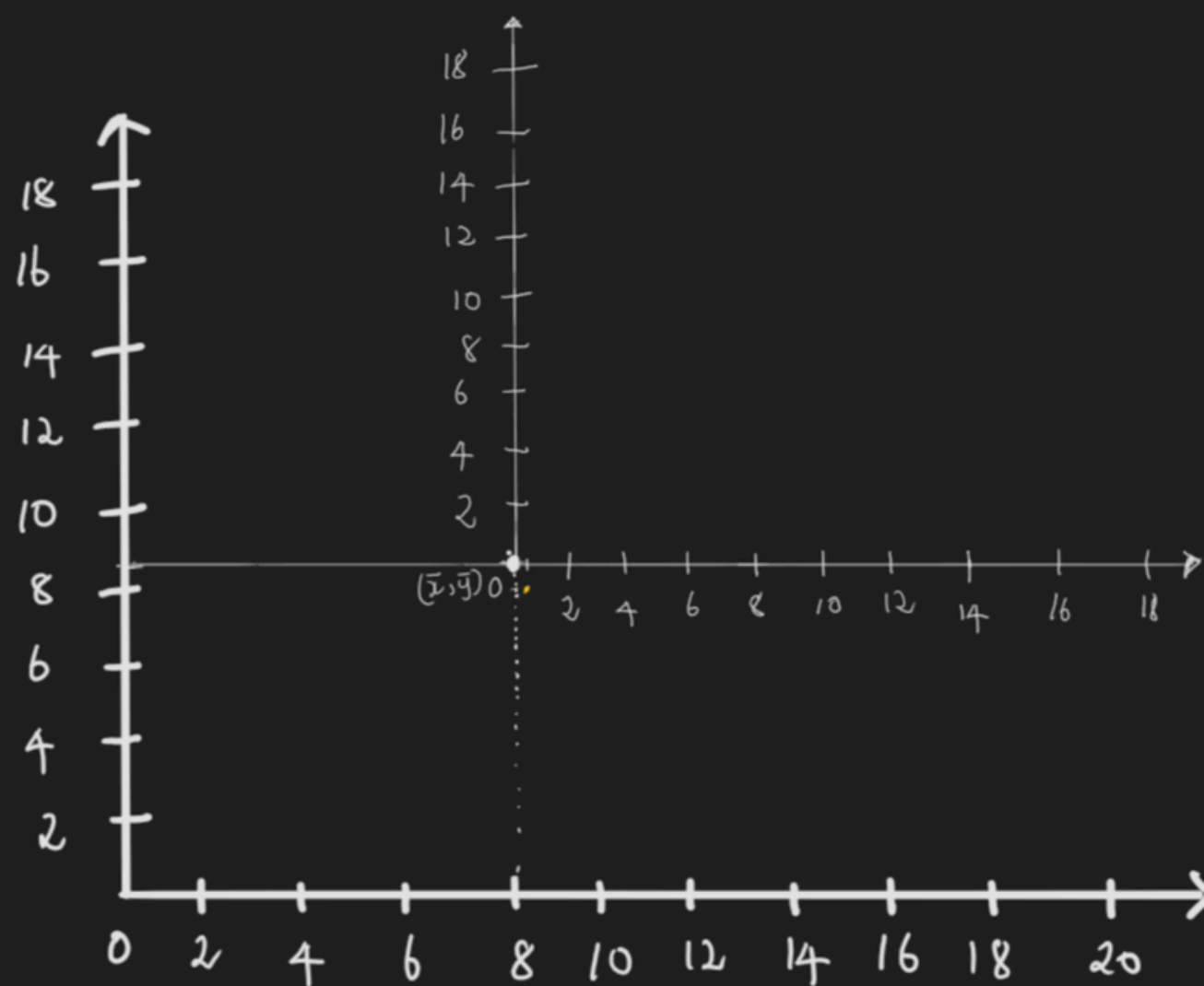


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$$\begin{bmatrix} 5.6128 \\ -5.1238 \end{bmatrix}$$

Now, let's draw a Coordinate system for Principal Components



① wkt, $\bar{x} = 8$, $\bar{y} = 8.5$.

Keep this point as a base
and reconstruct the
graph

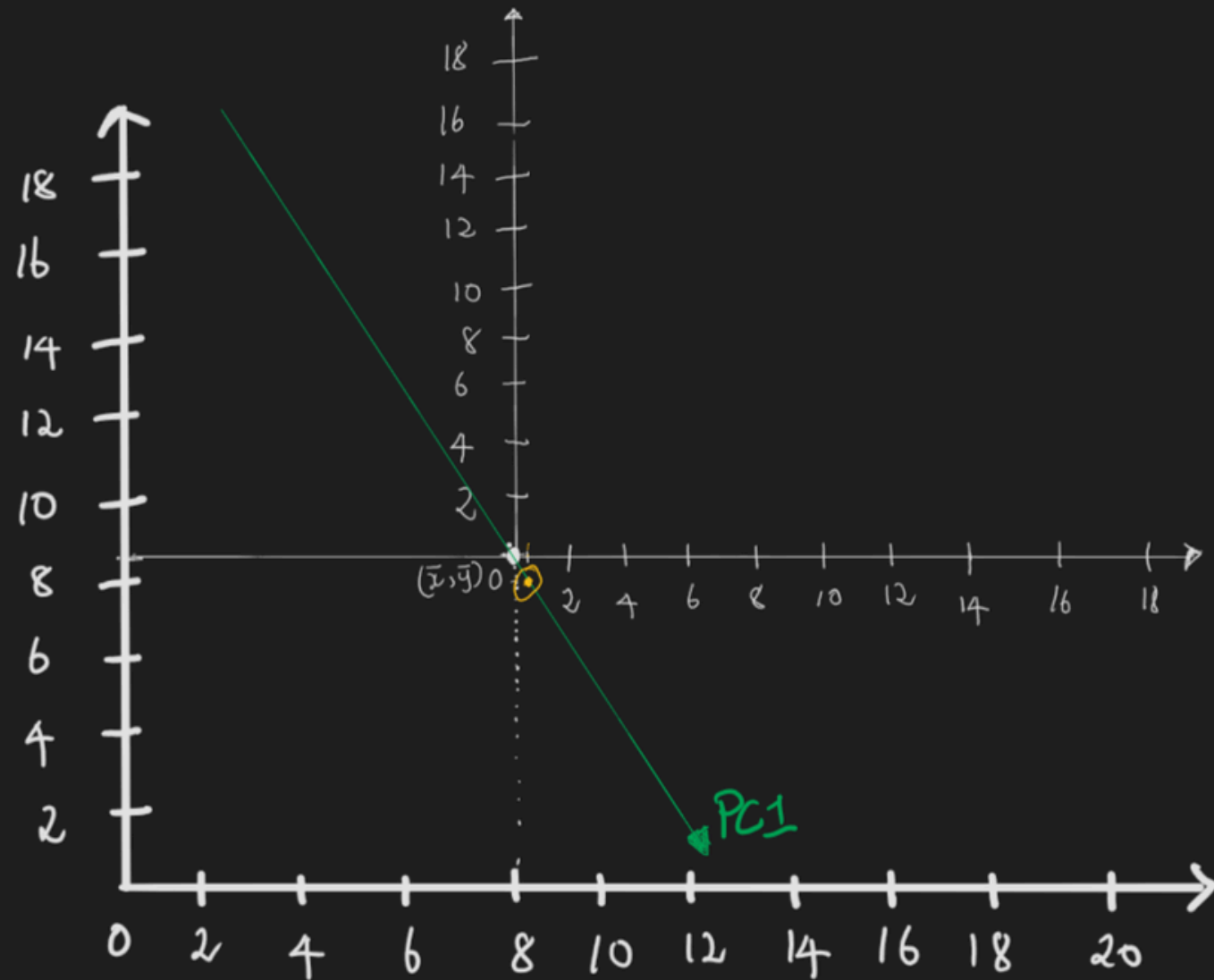
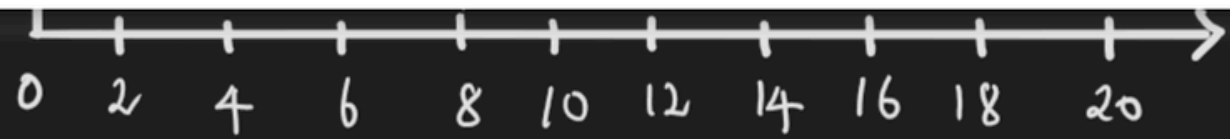
② Draw e_1 and e_2 i.e. eigen vectors.

$$\text{wkt, } e_1 = \begin{bmatrix} 0.5574 \\ -0.8303 \end{bmatrix} \rightarrow \text{x axis}$$

$$e_2 = \begin{bmatrix} 0.8303 \\ 0.5574 \end{bmatrix}$$

③ Connect $(\bar{x}, \bar{y})$ and e_1 which is the
eigen vector, to get the axis /

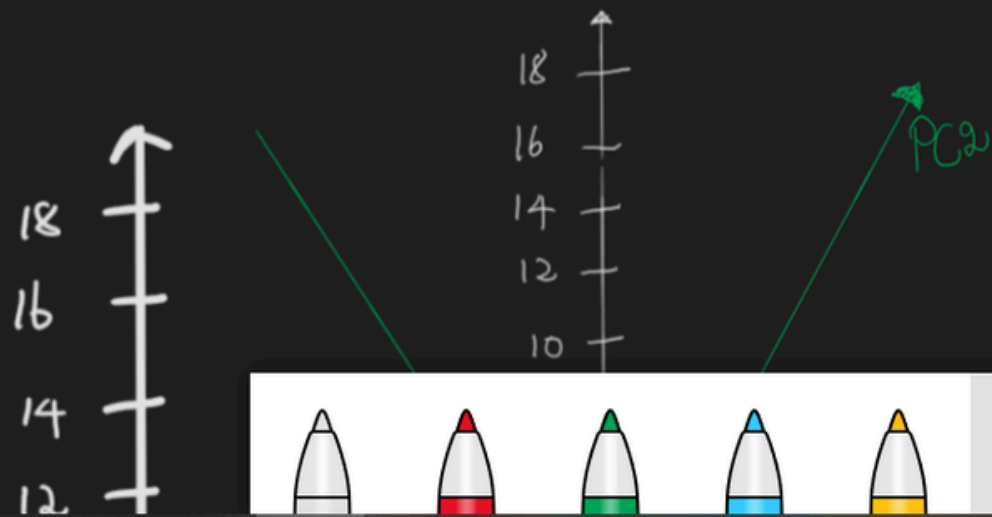




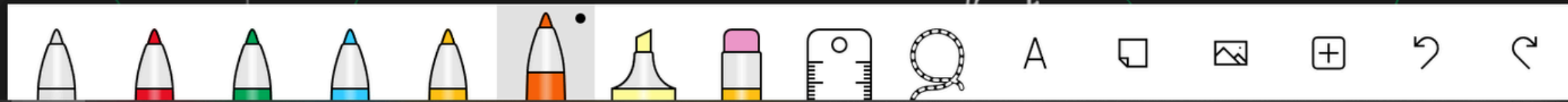
③ Connect $(\bar{x}, \bar{y})$ and e_1 which is the eigen vector, to get the axis / direction of PC1.

④ Connect $(\bar{x}, \bar{y})$ and e_2 to get the direction of PC2.

⑤ Erase all the other axis and mark the points on e_1 and e_2 .



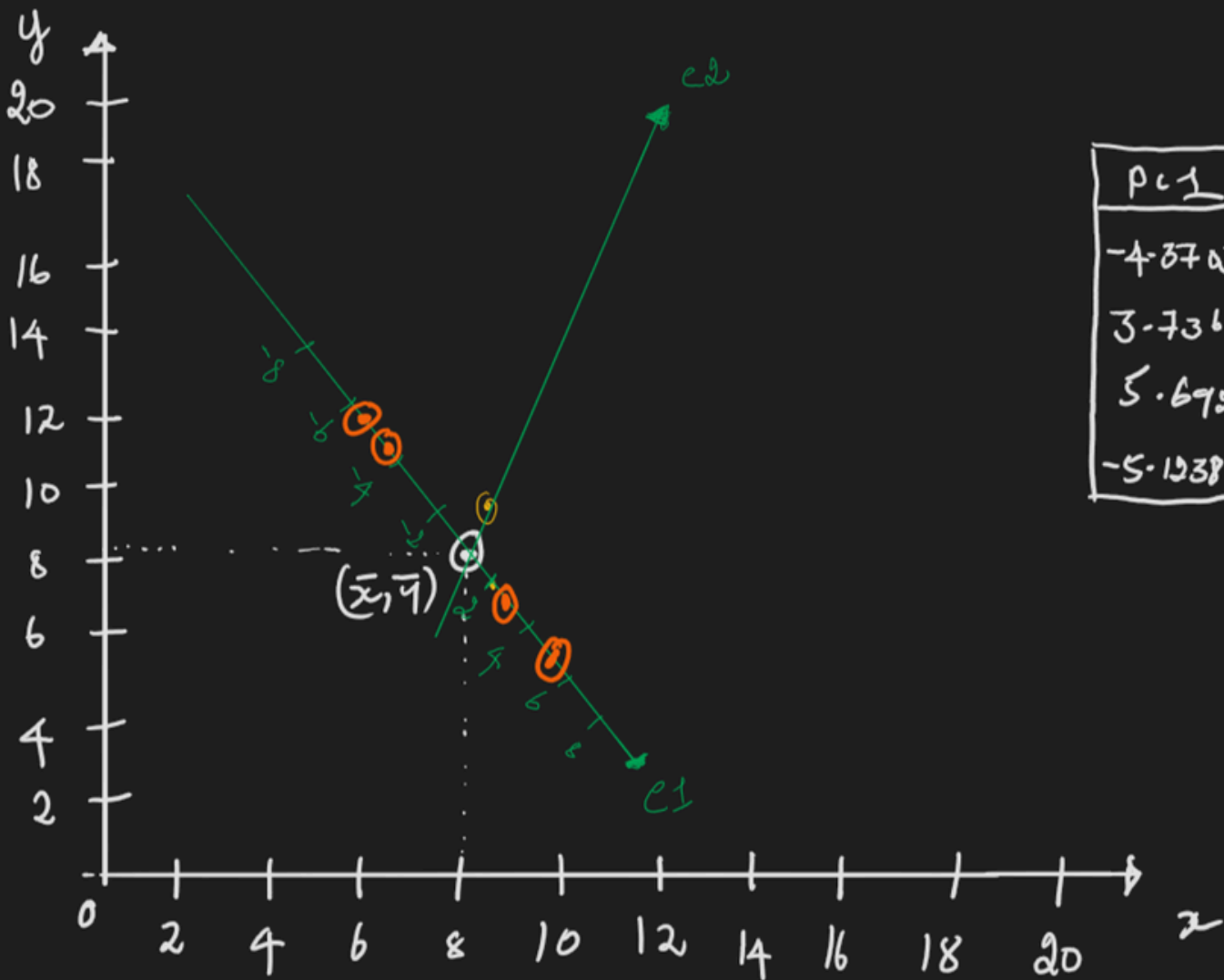
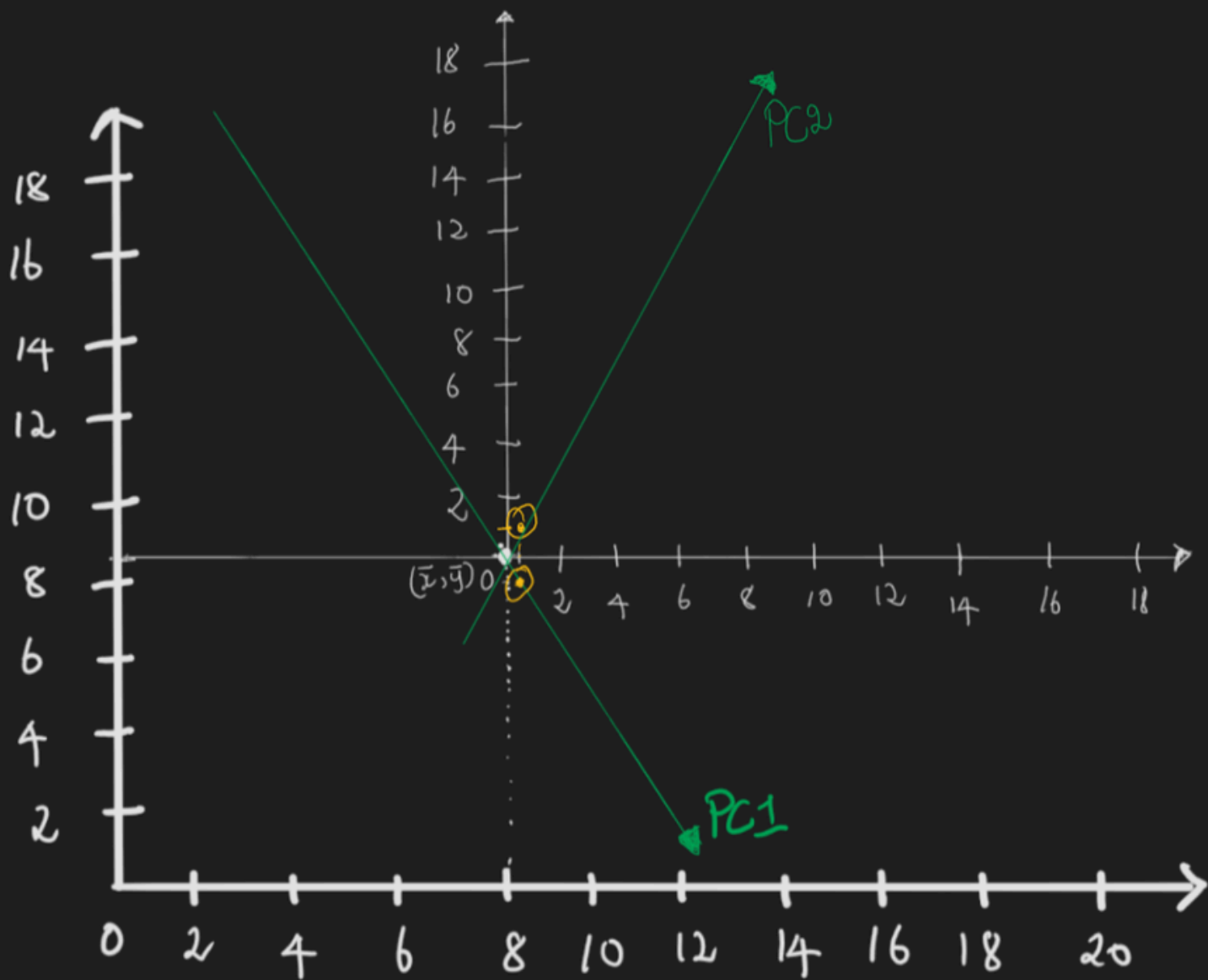
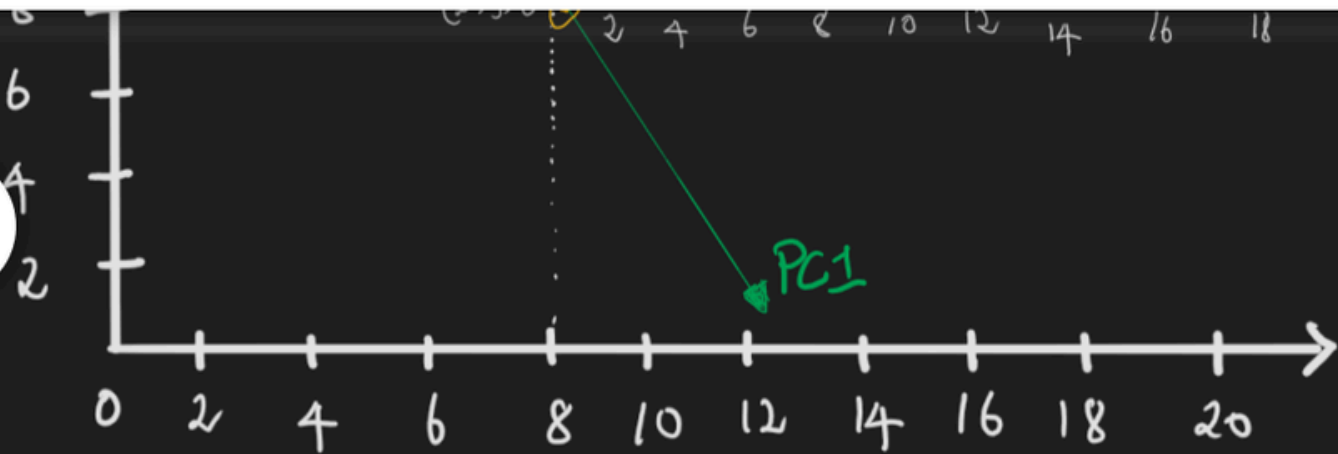
PC1
-4.8703
3.736
...





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PC1
-4.3703
3.736
5.6791
-5.1538

That's how

